Tutorial 10 (Friday)

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Problem #1: Find the linearization of the function $f(x,y) = \sqrt{x^2+16y^2}$ at the point $(3,1)$, and use it to approximate $f(2.9, 1.1)$.

Linear approximation: $L(x,y) \doteq f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$

$f(3,1) = \sqrt{9+16} = 5$ $\quad f_x(x,y) = \left(\frac{1}{2}\right)(x^2+16y^2)^{-1/2}(2x)$

$f_y(3,1) = \left(\frac{1}{2}\right)(5)^{-1}(6) = \frac{6}{10} = \frac{3}{5}$

$f_y(x,y) = \left(\frac{1}{2}\right)(x^2+16y^2)^{-1/2}(32y) = 16y\,(x^2+16y^2)^{-1/2}$

$f_y(3,1) = 16\,(5)^{-1} = \frac{16}{5}$

Thus, $L(2.9, 1.1) = 5 + \frac{3}{5}(2.9-3) + \frac{16}{5}(1.1-1)$

$= 5 + \frac{3}{5}(-0.1) + \frac{16}{5}(0.1)$

$= \boxed{5 - \frac{3}{5}\left(\frac{1}{10}\right) + \frac{16}{5}\left(\frac{1}{10}\right)}$

Problem #2: The length, width, and height of a box are measured as 3 ft, 7 ft, and 6 ft, respectively, with an error in measurement of at most 0.3 ft in each. Use differentials to estimate the maximum error (in feet) in the calculated volume of the box.

$x \quad y \quad z$

$|\Delta x| \le 0.3 \quad |\Delta y| \le 0.3 \quad |\Delta z| \le 0.3$

Volume: $V = xyz$ $\qquad$ use $dx = 0.3$, $dy = 0.3$, $dz = 0.3$

$dV = \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz$

$= yz\,dx + xz\,dy + xy\,dz$

$= (7)(6)(0.3) + (3)(6)(0.3) + (3)(7)(0.3)$

$= \frac{3}{10}[42 + 18 + 21]$

$= \frac{3}{10}[81]$

Problem #3: Suppose that f is a differentiable function with $f(7,0) = 6$ and $f_u(7,0) = 6$.
Let $w(u,v) = f(x(u,v), y(u,v))$ where $x = 3\cos u + 7\sin v$ and $y = 7\cos u \sin v$.
Find $w_u(\pi/2, \pi/2)$.

find $\frac{\partial x}{\partial u} = -3\sin u$

$\frac{\partial y}{\partial u} = -7\sin(u)\sin(v)$

So, $\frac{\partial x}{\partial u} = -3\sin(\pi/2) = -3$

$\frac{\partial y}{\partial u} = -7\underbrace{\sin(\pi/2)}_{=1}\underbrace{\sin(\pi/2)}_{=1} = -7$

$w_u = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u}$

note:

$x(\pi/2, \pi/2) = 3\cos(\pi/2) + 7\sin(\pi/2)$

$= 0 + 7 = 7$

$y(\pi/2, \pi/2) = 7\underbrace{\cos(\pi/2)}_{=0}\sin(\pi/2)$

$= 0$

So, $w_u(\pi/2, \pi/2) = f_u(7,0)$

Thus, $= 6\cdot(-3) + (8)(-7)$

$= -18 - 56$

$= -74$

Problem #4: Suppose that f is a twice differentiable function and that its second partial derivatives are continuous.

Let $h(t) = f(x(t), y(t))$ where $x = 3e^t$ and $y = t$.
Suppose that $f_x(3,0) = \sqrt{5}$, $f_y(3,0) = 4$, $f_{xx}(3,0) = 3$, $f_{xy}(3,0) = 1$, and $f_{yy}(3,0) = 4$.

Find $\frac{d^2h}{dt^2}$ when $t = 0$.

$t = 0$: $\quad x = 3e^0 = 3$, $\quad y = 0$

$\frac{dx}{dt} = 3e^t$ $\qquad \frac{d^2x}{dt^2} = 3e^t$

$\frac{dy}{dt} = 1$ $\qquad \frac{d^2y}{dt^2} = 0$

$h(t) = f(x(t), y(t))$

$\frac{dh}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$

$\frac{d^2h}{dt^2} = \frac{d}{dt}\left(\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}\right)$

$= \frac{\partial f}{\partial x}\frac{d^2x}{dt^2} + \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)\frac{dx}{dt}\frac{dx}{dt} + \frac{\partial^2 f}{\partial t\,\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial y}\frac{d^2y}{dt^2}$ $\quad (*)$

note: $\frac{\partial^2 f}{\partial t\,\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)\frac{dx}{dt} + \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)\frac{dy}{dt}$

$= \frac{\partial^2 f}{\partial x^2}\left(\frac{dx}{dt}\right) + \frac{\partial^2 f}{\partial y\,\partial x}\frac{dy}{dt}$

$= (3)(3e^t) + (4)(1)$

$t = 0$: $\quad = 9 + 4 = 13$

$\frac{\partial^2 f}{\partial t\,\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)\frac{dx}{dt} + \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)\frac{dy}{dt}$

$= \frac{\partial^2 f}{\partial x\,\partial y}\frac{dx}{dt} + \frac{\partial^2 f}{\partial y^2}\frac{dy}{dt}$

$= (4)(3e^t) + (1)(1)$

$t = 0$: $\quad = 12 + 1 = 13$

then $(*) = 13(3) + (3)(3) + (13)(1) + (4)(0)$

$= 39 + 9 + 13 = 61$ $\quad$ (so annoying...)

Problem #5: Find the directional derivative of the function $f(x,y) = \tan^{-1}(xy)$ at the point $(1,3)$ in the direction of the unit vector parallel to the vector $\mathbf{v} = 4\mathbf{i} + 5\mathbf{j}$.

$\nabla f(x,y) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} = \frac{y}{1+(xy)^2}\mathbf{i} + \frac{x}{1+(xy)^2}\mathbf{j}$

$\frac{\partial f}{\partial x} = y\frac{1}{1+(xy)^2}$ $\qquad \frac{\partial f}{\partial y} = x\frac{1}{1+(xy)^2}$

$\nabla f(1,3) = \frac{3}{1+9}\mathbf{i} + \frac{1}{1+9}\mathbf{j} = \frac{3}{10}\mathbf{i} + \frac{1}{10}\mathbf{j}$

$\|\mathbf{v}\| = \sqrt{4^2+5^2} = \sqrt{16+25} = \sqrt{41}$

unit vector direction: $\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{4}{\sqrt{41}}\mathbf{i} + \frac{5}{\sqrt{41}}\mathbf{j}$

$D_{\mathbf{u}}f(1,3) = \nabla f(1,3)\cdot\mathbf{u}$

$= \left(\frac{3}{10}\mathbf{i} + \frac{1}{10}\mathbf{j}\right)\cdot\left(\frac{4}{\sqrt{41}}\mathbf{i} + \frac{5}{\sqrt{41}}\mathbf{j}\right)$

$= \frac{12}{10\sqrt{41}} + \frac{5}{10\sqrt{41}} = \frac{17}{10\sqrt{41}}$

Problem #6: The temperature at a point (x, y, z) is given by

$T(x,y,z) = 10e^{-x^2-3y^2-2z^2}$.

In which direction does the temperature increase fastest at the point $(4, 3, 1)$?
Express your answer as a UNIT vector.

Thm: if f is diffble of 2-3 variables then the max value of the directional derivative $D_{\mathbf{u}}f(x)$ is $|\nabla f(x)|$ and occurs when $\mathbf{u}$ has the same direction as $\nabla f(x)$.

Thus $\nabla T = \frac{\partial T}{\partial x}\mathbf{i} + \frac{\partial T}{\partial y}\mathbf{j} + \frac{\partial T}{\partial z}\mathbf{k}$ pt $(4,3,1)$:

$\frac{\partial T}{\partial x} = 10e^{-x^2-3y^2-2z^2}(-2x) = 10e^{-45}(-8) = -80e^{-45}$

$\frac{\partial T}{\partial y} = 10e^{-x^2-3y^2-2z^2}(-6y) = 10e^{-45}(-18) = -180e^{-45}$

$\frac{\partial T}{\partial z} = 10e^{-x^2-3y^2-2z^2}(-4z) = 10e^{-45}(-4) = -40e^{-45}$

Aside: $10e^{-16-3(9)-2(1)} = 10e^{-45}$

$\nabla T(4,3,1) = -80e^{-45}\mathbf{i} - 180e^{-45}\mathbf{j} - 40e^{-45}\mathbf{k}$

$= 20e^{-45}(-4\mathbf{i} - 9\mathbf{j} - 2\mathbf{k})$

$\mathbf{v} = -4\mathbf{i} - 9\mathbf{j} - 2\mathbf{k}$

$\|\mathbf{v}\| = \sqrt{16+81+4} = \sqrt{101}$

then by thm above, direction is:

$\frac{-4}{\sqrt{101}}\mathbf{i} - \frac{9}{\sqrt{101}}\mathbf{j} - \frac{2}{\sqrt{101}}\mathbf{k}$ use: $\boxed{\frac{-4}{\sqrt{101}}, \frac{-9}{\sqrt{101}}, \frac{-2}{\sqrt{101}}}$

Problem #7: Let $f(x,y) = xe^{xy}$. Find the maximum rate of change of f at the point $(-2, -2)$.

Again, use above thm.

$\nabla f(x,y) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} = (e^{xy} + xye^{xy})\mathbf{i} + x^2e^{xy}\mathbf{j}$

$= (e^4 + 4e^4)\mathbf{i} + 4e^4\mathbf{j}$

$|\nabla f(-2,-2)| = \sqrt{(e^4+4e^4)^2 + 16e^8}$

$\frac{\partial f}{\partial x} = e^{xy} + xe^{xy}(y)$

$\frac{\partial f}{\partial y} = xe^{xy}(x)$

if you can simplify: $\sqrt{25e^8 + 16e^8} = \sqrt{41}\,e^4$

ok for childsmath.

YAY NO MORE ASSIGNMENTS HAHAHAHAHAHA